



Editorial

SAPFLUXNET: towards a global database of sap flow measurements

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Plant transpiration is the main evaporative flux from terrestrial ecosystems; it controls land surface energy balance, determines catchment hydrological responses and influences regional and global climate. Transpiration regulation by plants is a key (and still not completely understood) process that underlies vegetation drought responses and land evaporative fluxes under global change scenarios. Thermometric methods of sap flow measurement have now been widely used to quantify whole-plant and stand transpiration in forests, shrublands and orchards around the world. A large body of research has applied sap flow methods to analyse seasonal and diurnal patterns of transpiration and to quantify their responses to hydroclimatic variability, but syntheses of sap flow data at regional to global scales are extremely rare. Here we present the SAPFLUXNET initiative, aimed at building the first global database of plant-level sap flow measurements. A preliminary metadata survey launched in December 2015 showed an encouraging response by the sap flow community, with sap flow data sets from field studies representing >160 species and >120 globally distributed sites. The main goal of SAPFLUXNET is to analyse the ecological factors driving plant- and stand-level transpiration. SAPFLUXNET will open promising research avenues at an unprecedented global scope, namely: (i) exploring the spatio-temporal variability of plant transpiration and its relationship with plant and stand attributes, (ii) summarizing physiological regulation of transpiration by means of few water-use traits, usable for land surface models, (iii) improving our understanding of the coordination between gas exchange and plant-level traits (e.g., hydraulics) and (iv) analysing the ecological factors controlling stand transpiration and evapotranspiration partitioning. Finally, SAPFLUXNET can provide a benchmark to test models of physiological controls of transpiration, contributing to improve the accuracy of individual water stress responses, a key element to obtain robust predictions of vegetation responses to climate change.

Keywords: drought, evapotranspiration, functional traits, plant hydraulics, plant water use, sap flux, transpiration.

Plant transpiration is a key component of land evaporative fluxes, but it is hard to measure

Plant transpiration is the main evaporative flux from terrestrial ecosystems (Schlesinger and Jasechko 2014); it controls land surface energy balance, determines catchment hydrological responses and influences regional and global climate (Bonan 2008). Because of the central role played by their water economy, plants adjust physiologically and structurally to maintain their hydraulic integrity and metabolic function under drought at various time and organizational scales (Maseda and Fernández 2006).

These adjustments determine how plants regulate transpiration under fluctuating conditions of water supply and demand, resulting in multiple water-use strategies within and across ecosystems. The diversity of water-use strategies underlies plant drought responses, and these responses are major drivers of demographic rates (McDowell et al. 2008), community dynamics (Silvertown et al. 2015) and ecosystem function. Transpiration regulation by plants is thus a key (and still not completely understood) process that underlies vegetation drought responses (McDowell et al. 2013) and land evaporative fluxes (Oishi et al. 2010, Miralles et al. 2011) under global change scenarios.

The measurement of plant transpiration and its drivers is deeply rooted in the history of quantitative experiments in plant biology but most of the experimental techniques developed before the 1950s were intrusive and unsuitable for the investigation of the driving factors of transpiration of living plants, especially under field conditions (Wullschlegel et al. 1998). The advent of gas exchange systems allowed the development of automated cuvettes to measure transpiration from leaves to whole plants, but the upscaling of leaf-to-plant transpiration is highly labour-intensive and error-prone, while chamber-based measurements of whole-plant transpiration are logistically demanding and can alter canopy microclimate (Goulden and Field 1994). Micrometeorological techniques, stand or catchment water balances and remote sensing approaches provide evapotranspiration estimates, but neither of these methods is suitable to study water-use strategies of individual plants across the entire spatio-temporal domain.

Sap flow methods: development, limitations and contribution to progress in understanding tree and ecosystem physiology

Thermometric methods of sap flow measurement were devised during the 1930s and refined in the late 1950s (Marshall 1958). The main methodological families were already established by the mid-1980s and many improvements and new derivations have continued to appear (Čermák et al. 2004, Vandegehuchte and Steppe 2013 and references therein). Sap flow methods measure either total sap flow through a plant's organ or sap flux density, the flow rate per unit conducting area (Vandegehuchte and Steppe 2013). They use pulsed or continuous heating and track temperature changes caused by convective heat transport by moving sap in different plant organs along the transpiration pathway (Goldstein et al. 1998, Burgess et al. 2000). Given the easy replication and automation of sap flow probes, they have been widely used on main, woody stems to quantify tree and stand transpiration (Granier et al. 1996).

Each of the sap flow methods has distinct advantages and methodological constraints. No single method is able to measure the entire range of sap flow rates and all of them are subject to one or more of the following issues (Vandegehuchte and Steppe 2013): incomplete contact of the probe with the sapwood, natural temperature gradients, the need for species-specific calibrations and parameters, uncertainty in baseline flow estimates, wounding effects and sensor drift. These potential artefacts may be aggravated by within-method differences in sensor design among manufacturers and laboratories (Lu et al. 2004). Unsurprisingly, validation studies have raised concerns about the potential biases incurred by most sap flow methods (Steppe et al. 2010). Additional uncertainty results from upscaling from sensors to trees and to stands (Oren et al. 1998, Čermák et al. 2004). Nevertheless, all methods yield reasonable temporal

patterns of plant transpiration and sap flow-derived transpiration has often compared well with independent measurements (McCulloh et al. 2007) and with evapotranspiration at larger scales (Wilson et al. 2001).

Sap flow methods have greatly contributed to addressing fundamental questions of (mostly woody) plant and ecosystem physiology, such as age/height- (Hubbard et al. 1999) and drought-related variations in the water transport capacity of plants (Cochard et al. 1996), the mechanisms of stem water storage dynamics (Goldstein et al. 1998) or foliar water uptake (Burgess and Dawson 2004). Sap flow methods have also been successfully applied to study broader ecological questions, such as the effects of tree diversity on ecosystem functioning (Forrester et al. 2010). At a more applied level, sap flow is frequently measured in studies designed to guide silvicultural or agronomic practices (Jones 2004, Aranda et al. 2012). A large body of research has applied sap flow methods to analyse seasonal and diurnal patterns of transpiration and canopy (stomatal) conductance (G_s) (Phillips and Oren 1998) and to quantify their responses to hydroclimatic variability.

The time is ripe for a global database of sap flow measurements: the SAPFLUXNET initiative

Sap flow probes are relatively affordable and easy to deploy in the field, but they have logistic requirements that preclude the long-term maintenance of multiple experimental sites across broad ecological gradients. Data syntheses of sap flow (or G_s) responses to hydroclimatic factors across species have either focused on responses to single variables (Oren et al. 1999) or they have been limited to a few closely related species (Duursma et al. 2008), likely reflecting limitations in data availability in relatively homogeneous and comparable formats. Indeed, a recent global synthesis of sap flow focusing on pine stands' response to variation in leaf area index, soil and climate variables, required digitizing published data (Torngern et al. 2016), introducing additional, unnecessary errors. Only a few studies have analysed responses of sap flow and/or G_s across a diverse group of tree species (Granier et al. 2000) or along wide climatic gradients within a species (Poyatos et al. 2007). Other quantitative syntheses have typically analysed maximum values of sap flow or G_s , either measured during short periods or retrieved from published sources. These studies have mostly dealt with how maximum sap flow rates vary with allometry (Meinzer et al. 2005), plant hydraulics (Manzoni et al. 2013) or wood traits (Kallarackal et al. 2013).

Increasing data sharing and reuse promotes progress in ecology by extending the value of data beyond individual studies (Wolkovich et al. 2012). Global and regional networks of biosphere-atmosphere carbon and water exchange curate regional and global data sets, improve data interoperability and enable collaborations and data syntheses (Papale et al. 2012). Recent

methodological progress in data management, database design (Boden et al. 2013) and open-source software tools have occurred in concert with the increasing availability of global databases. Functional and ecosystem ecologists are already producing global syntheses on vegetation controls of water balance and on plant's drought vulnerability (Choat et al. 2012, Williams et al. 2012). However, the sap flow community has not yet fully embraced this challenge; for instance, sap flow data sets are mostly absent from the main data repositories for ecological or Earth system sciences (one data set for canopy transpiration in try-db.org; one data set, with peak values only, in datadryad.org; none in pangaea.de, re3data.org or zenodo.org; all repositories accessed May 2016).

The sap flow-related literature is already substantial: a Scopus search ('sap flow' OR 'sapflow' OR 'sap flux' OR 'sapflux' in title, abstract or keywords) returned >2500 papers, whose keywords included the scientific names of >250 plant species, according to The Plant List (2013, Version 1.1.; <http://www.theplantlist.org/>). These figures likely overestimate the number of potential data sources, but they nevertheless suggest that an ecologically relevant compilation is feasible. We searched the title, abstract and keywords of these papers to retrieve the sap flow method(s) used in each paper, and we show that the heat dissipation method (Granier 1985) is the most popular, by far (Figure 1). The SAPFLUXNET initiative (<http://sapfluxnet.creaf.cat/>) is building the first global database of plant-level sap flow measurements (obtained using thermometric methods) to analyse the environmental and physiological factors driving tree- and stand-level

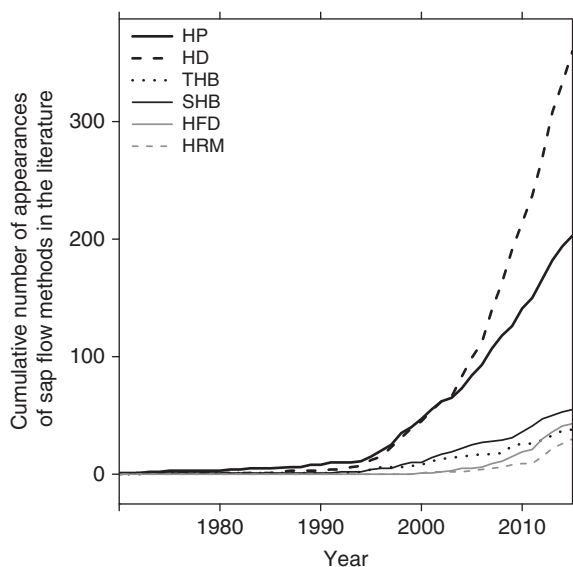


Figure 1. Cumulative count of appearances of different sap flow methods in a Scopus search of sap flow-related literature ('sap flow' OR 'sapflow' OR 'sap flux' OR 'sapflux' in title, abstract or keywords). Note that counts represent individual appearances of each method, not papers (e.g., a paper can have multiple methods). HP, heat pulse methods, including Compensation Heat Pulse, Tmax Method, Calibrated Average Gradient and Sapflow+; HD, heat dissipation; THB, tissue heat balance; SHB, stem heat balance; HFD, heat field deformation; HRM, heat ratio method.

transpiration. A preliminary metadata survey launched in December 2015 showed an encouraging response by the sap flow community. Potentially contributed sap flow data sets from field studies represent >160 species and >120 globally distributed sites (Figure 2). We have now started the data collection stage, and we welcome processed sap flow data sets measured at the stem or whole-plant level (but not at the branch level), under field conditions and at sub-daily intervals. Please see the SAPFLUXNET wiki <https://github.com/sapfluxnet/sapfluxnet-public/wiki> for further details on minimum requirements for sap flow, environmental and ancillary data. Our ultimate goal is that, by the end of the project (September 2018), a first version of the SAPFLUXNET database is made available to data contributors. After an embargo period, the database will be made publicly available to the scientific community.

Building a harmonized database from highly heterogeneous data sources is a challenging endeavour. We have designed a semi-automated, reproducible and version-controlled, data ingestion process using the tools provided by the R environment (e.g., R language for computing and statistics, interactive R Markdown documents and Shiny apps). Data incorporation to the server and initial data quality control (timestamp and units standardization, species and geographic coordinates validation, gap quantification) is automatically performed by R functions (Figure 3).

SAPFLUXNET will compile, curate and disseminate processed data. Since sap flow measurements are affected by a substantial methodological variability, sap flow time series in SAPFLUXNET are documented with metadata on sensor characteristics, data processing and upscaling procedures, provided by data contributors. For example, we specify whether species-specific calibrations (Steppe et al. 2010) or natural gradient corrections (Do and Rocheteau 2002) have been applied to the raw data. These metadata will allow us to account for variability caused by methodological issues in joint analyses using the database. SAPFLUXNET will thus allow a first quantification of the impacts of methodological variability on sap flow measurements across a large number of species. We expect that these analyses will foster the development of standardized protocols within the sap flow community, as a prerequisite towards a fully coherent database similar to what FLUXNET has achieved for ecosystem-level CO₂ and water fluxes (Papale et al. 2012). There have been recent advances in this direction with the dissemination of code for robust, semi-automatic flux baselining for heat dissipation probes (Oishi et al. 2016) and for generalized integration of radial variation of sap flow in the scaling to whole-tree water use (Berdanier et al. 2016).

Unravelling the ecological drivers of plant transpiration to understand drought vulnerability and ecosystem physiology

The SAPFLUXNET initiative will provide opportunities to improve our understanding of the ecological determinants of plant water

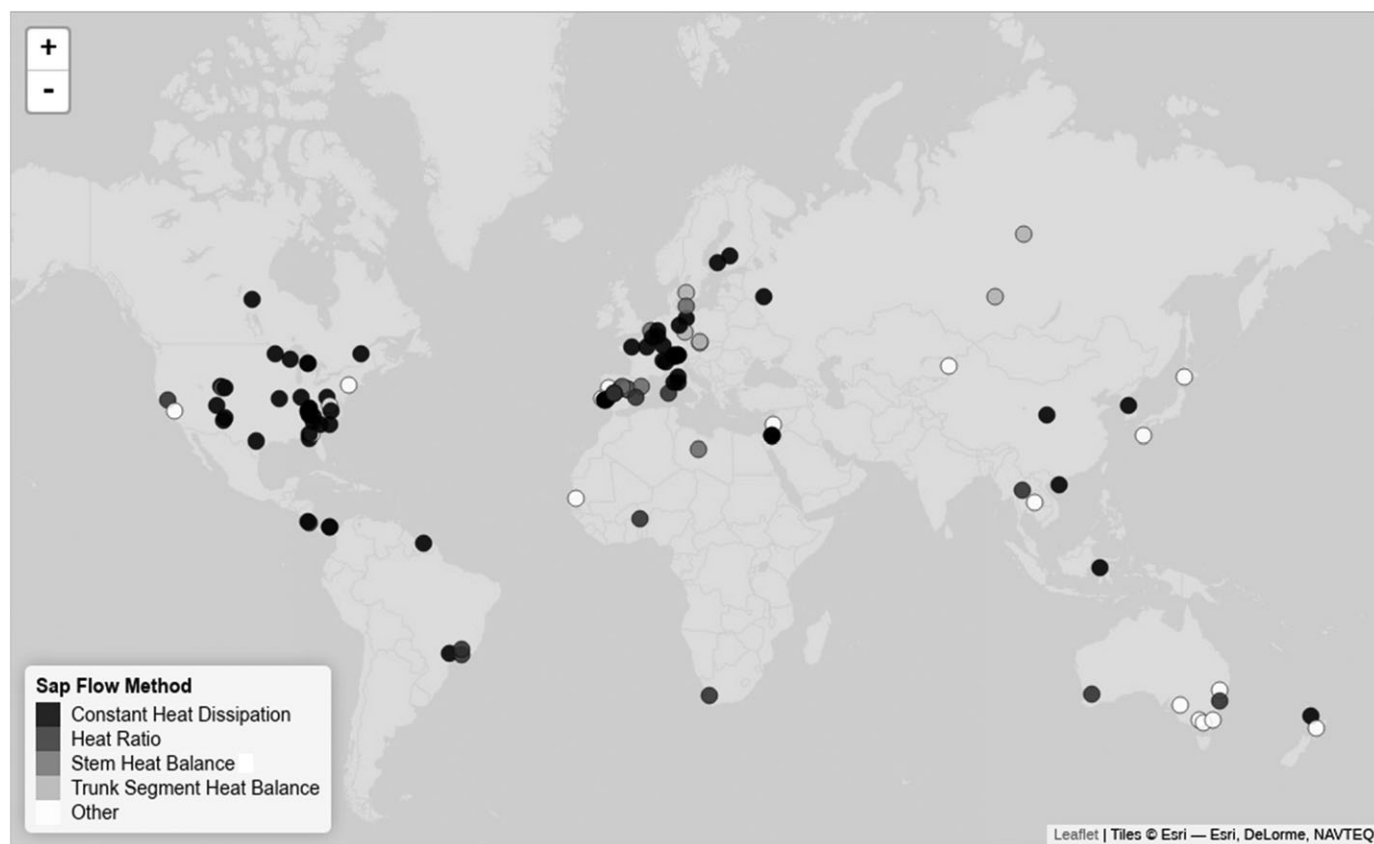


Figure 2. Geographic distribution of potential sap flow data sources, coded by sap flow method, based on a preliminary survey. An interactive summary of this preliminary survey, with further information on the growth condition of the stand, the number of species and trees can be found at http://sapflux-net.creaf.cat/shiny/preliminary_dashboard/.

use and drought responses. It will allow description of how annually or seasonally integrated plant transpiration vary across biomes, plant functional types or taxonomic groups. SAPFLUXNET will open promising research avenues at an unprecedented global scope, namely: (i) exploring the spatio-temporal variability of plant transpiration and its relationship with plant and stand attributes, (ii) summarizing physiological regulation of transpiration by means of water-use traits, representing maximum water transport rates and sensitivity to environmental factors, usable for land surface models, (iii) improving our understanding of the coordination between gas exchange and plant-level traits (e.g., hydraulics) and (iv) analysing the ecological factors controlling stand transpiration and evapotranspiration partitioning.

SAPFLUXNET will allow the first analyses of biogeographical and climate-driven patterns of plant transpiration. The variation in transpiration with water availability will be assessed using probabilistic, multi-scalar drought indices (Vicente-Serrano et al. 2010), which include drought responses at multiple temporal scales and allow comparisons across ecosystems with contrasting water availability. Moreover, SAPFLUXNET will extend scaling relationships between sap flux density and plant size and hydraulic architecture (e.g., sapwood area, leaf area), to test for functional convergence across functional types, genera or environmental gradients

(Zeppel 2013). As stand-level hydraulic adjustments (Whitehead et al. 1984) covary with plant water availability, a relevant question will be to assess how these scaling relationships vary with stand structure (e.g., basal area, density, canopy height).

One of the main goals of SAPFLUXNET will be to summarize physiological regulation of transpiration by means of few water-use traits. For example, semi-empirical responses of sap flow (or canopy conductance) to radiation, vapour pressure deficit and soil moisture can be embedded in dynamic, hierarchical Bayesian models, which can deal with observation, process and parameter uncertainties enabling the analysis of heterogeneous sap flow data sets (Ward et al. 2013). Water-use traits derived from these models represent complex ecophysiological strategies as emerging relationships, such as the trade-off between maximum stomatal conductance and sensitivity to vapour pressure deficit, which arises from stomatal limitation of xylem embolism (Oren et al. 1999). This trade-off provides a benchmark to investigate variation in physiological regulation of transpiration across biomes, climates (Poyatos et al. 2007), wood anatomy types (Meinzer et al. 2013) or taxonomic groups (Johnson et al. 2012). Moreover, physiological responses to soil moisture, radiation and atmospheric turbulence are virtually unexplored at regional to global scales.

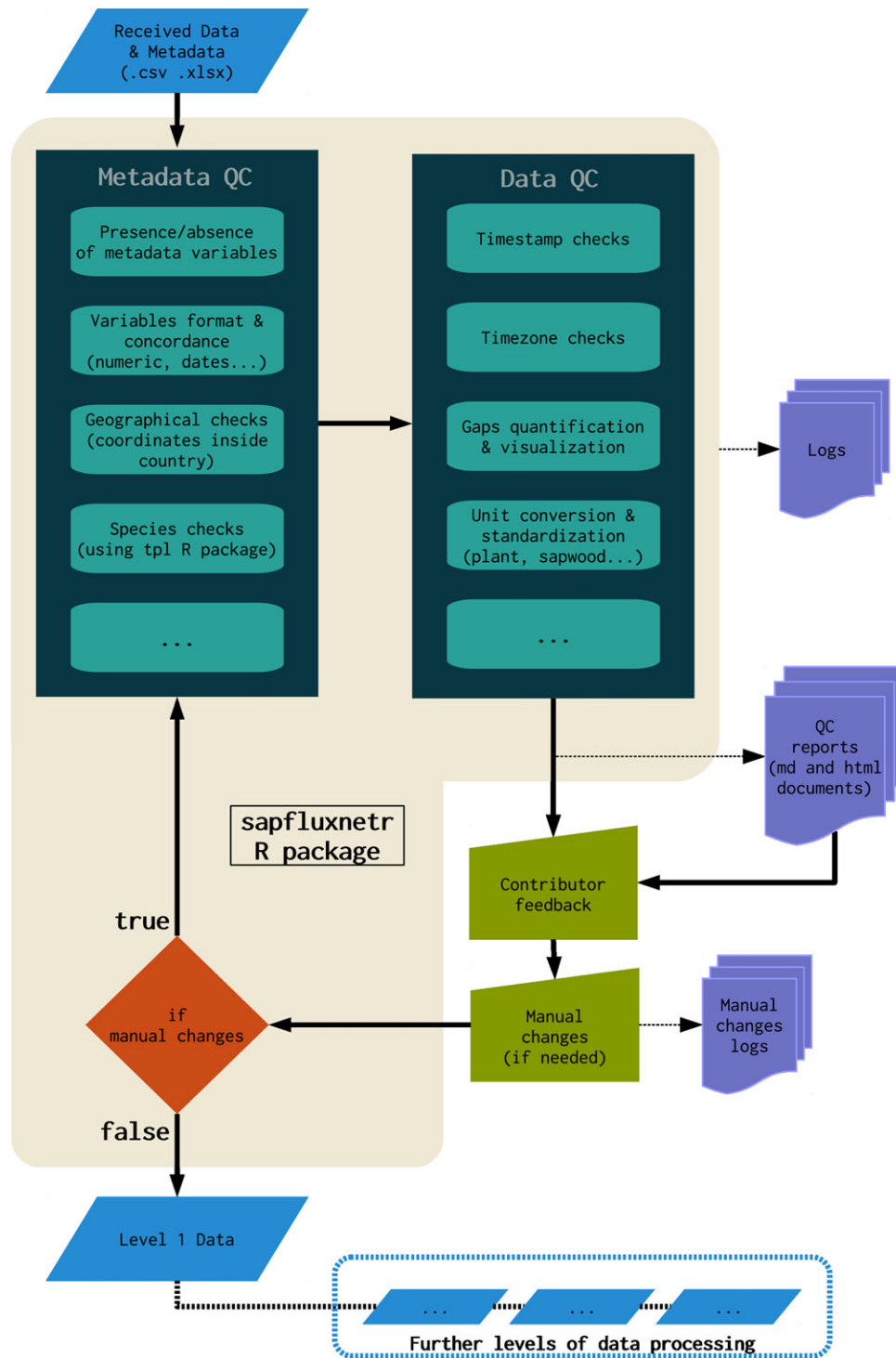


Figure 3. Data flow from reception to production of Level 1 data sets. Initial data and metadata quality control is encapsulated in an R package under continuous development. This semi-automated procedure generates interactive quality control (QC) reports that are shared with data contributors to resolve any issues with the data. All the individual steps in this procedure and any manual changes on the data sets are traceable because they are logged in structured text files. Beyond Level 1, data processing is under implementation, as depicted by discontinuous lines.

Coordination of canopy gas exchange and whole-plant hydraulics has long been proposed (Meinzer 2002), but only recently have global syntheses of hydraulic traits started to provide a broader picture of the ecological patterns and the multiple

facets of drought tolerance in plants. Whole-plant water-use traits derived from SAPFLUXNET will help complete the worldwide spectrum of hydraulic strategies defined by gas exchange and water potential regulation (Martínez-Vilalta et al. 2014),

hydraulic safety margins (Choat et al. 2012), capacitance (Meinzer et al. 2009) and leaf water relations (Bartlett et al. 2012). Combining global sap flow data with on-site water potentials will also simplify the parameterization of whole-plant transpiration models (Sperry and Love 2015).

At the ecosystem level, SAPFLUXNET can provide globally distributed values of seasonally and annually aggregated stand transpiration, which will improve our understanding of the biophysical controls of vegetation–atmosphere fluxes (Köstner 2001) and will help constrain the estimates of transpiration-to-evapotranspiration ratios (Schlesinger and Jasechko 2014). SAPFLUXNET can help to link plant functional traits with ecosystem water and carbon fluxes (Reichstein et al. 2014). Finally, SAPFLUXNET can provide a benchmark to test algorithms of physiological controls of transpiration embedded in process-based models (Williams et al. 2001), contributing to improve the accuracy of individual water stress responses, a key element to obtain robust predictions of vegetation responses to climate change (Levine et al. 2015).

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Conflict of interest

None declared.

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